

GROWMARK, INC.**1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION**

Product Name: Bullzeye™ 5.4 Plus
EPA Reg. No.: 103591-8-534
Product Type: Herbicide
Company Name: Growmark, Inc.
1705 Towanda Ave., Bloomington, IL 61701
Telephone Numbers: For Chemical Emergency, Spill, Leak, Fire, Exposure, or Accident, call CHEMTREC Day or Night: 1-800-424-9300
For Medical Emergencies Only, Call 1-877-325-1840

This product is an EPA FIFRA registered pesticide. Some classifications on this SDS are not the same as the FIFRA label. Certain sections of this SDS are superseded by federal law governed by EPA for a registered pesticide. Please see section 15. REGULATORY INFORMATION for explanation.

2. HAZARD IDENTIFICATION

Harmful to aquatic life (H402)

HAZARD CLASSIFICATION

Health Hazards	Category	Physical Hazards	Category
None	--	None	-
		Environmental Hazards	Category
		Hazardous to the aquatic environment, short-term (acute)	3

HAZARDS NOT REQUIRING CLASSIFICATION

DO NOT MIX, STORE OR APPLY THIS PRODUCT OR SPRAY SOLUTIONS OF THIS PRODUCT IN GALVANIZED STEEL OR UNLINED STEEL (EXCEPT STAINLESS STEEL) CONTAINERS OR SPRAY TANKS. This product or spray solutions of this product react with such containers and tanks to produce hydrogen gas which may form a highly combustible gas mixture. This gas mixture could flash or explode, causing serious personal injury, if ignited by open flame, spark, welder's torch, lighted cigarette or other ignition source.

PRECAUTIONARY STATEMENTS

Avoid release to the environment not in accordance with the product label. (P273)

Dispose of contents / container in accordance with local regulations. Refer to the product label for specific disposal instructions. (P501)

3. COMPOSITION / INFORMATION ON INGREDIENTS

<u>Common Name</u>	<u>Chemical Name</u>	<u>CAS #</u>	<u>Composition</u>
Glyphosate IPA Salt	N-(phosphonomethyl)glycine, isopropylamine salt	38641-94-0	53.8%

NOTE: Ingredients not precisely identified are proprietary or non-hazardous. Values are not product specifications.

4. FIRST AID MEASURES

Have the product container or label with you when calling a poison control center or doctor, or going for treatment. IN CASE OF EMERGENCY INVOLVING THIS HERBICIDE PRODUCT, OR FOR MEDICAL ASSISTANCE, CALL (800) 424-9300.

5. FIREFIGHTING MEASURES

Flash Point (PM): Does not flash

Flammable Limits: Not determined

Fire and Explosion Hazards: None known.

Extinguishing Medium: Water spray, foam, dry chemical, or CO₂. Minimize the use of water to prevent environmental contamination.

Fire Fighting Equipment: Firefighters should be equipped with self-contained positive pressure breathing apparatus and full bunker gear.

Fire Fighting Instructions: Isolate hazard area and keep unnecessary persons from entering. Dike and contain firefighting water to prevent runoff and environmental contamination.

Hazardous Combustion Products: Oxides of carbon, nitrogen, and phosphorus.

NFPA Ratings: Health – 0 / Flammability – 1 / Reactivity - 1

6. ACCIDENTAL RELEASE MEASURES

Spills: Wear appropriate protective gear as described in Section 8 of this document. Dike spill using absorbent or impervious materials such as earth, sand or clay. Collect and contain contaminated absorbent and dike material for disposal. Pump any free liquid into an appropriate closed container. Collect washings for disposal. Decontaminate tools and equipment following cleanup. Large spills may be reportable to the National Response Center (800-424-8802) and to state and/or local agencies.

Environmental Precautions: Prevent material from entering public sewer systems or any waterways. Do not flush to drain. Large spills to soil or similar surfaces may necessitate removal of topsoil. The affected area should be removed and placed in an appropriate container for disposal.

7. HANDLING AND STORAGE

Handling: Remove and wash contaminated clothing before reuse.

Storage: Keep container closed to prevent spills and contamination. Store above 5°F (-15°C) to keep product from crystallizing. Crystals will settle to the bottom. If allowed to crystallize, place in a warm room 68°F (20°C) for several days to redissolve and roll or shake container to mix well before using.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Engineering Controls: Facilities storing or utilizing this material should be equipped with an eyewash station and a safety shower. Ensure adequate ventilation.

Personal Protective Equipment: Applicators and other handlers must wear: long-sleeved shirt and long pants, and shoes plus socks.

General: Follow manufacturer's instructions for cleaning/maintaining Personal Protective Equipment (PPE). If there are no such instructions for washables, use detergent and hot water. Keep and wash PPE separately from other laundry.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	colorless – amber liquid	Solubility:	water: completely miscible
Odor:	odorless	Partition coefficient:	log P _{ow} : <-3.2 @ 25°C (glyphosate)
Melting/freezing point:	not determined	Vapor pressure:	no significant volatility
Boiling point/Boiling range:	not determined	Density:	1.206 g/cm ³ @ 20°C
Flammability:	not flammable	Relative vapor density:	not determined
Flammability limits (upper/lower):	not determined	Particle characteristics:	not available
Flash point:	does not flash	Explosive Properties:	not available
Auto-ignition temperature:	not determined		
Decomposition temperature:	not determined		
pH:	4.6-4.8		
Kinematic viscosity:	not determined		

10. STABILITY AND REACTIVITY

CONDITIONS TO AVOID / HAZARDOUS REACTIONS: Reacts with galvanized steel or unlined mild steel to produce hydrogen, a highly flammable gas that could explode.

CHEMICAL STABILITY: This product is stable under normal handling and storage conditions.

INCOMPATIBILITY WITH OTHER MATERIALS: Galvanized steel, unlined mild steel

HAZARDOUS DECOMPOSITION PRODUCTS: Thermal decomposition can produce oxides of carbon, nitrogen, and phosphorus.

HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL INFORMATION

ORAL TOXICITY (rat LD₅₀): > 5,000 mg/kg

DERMAL TOXICITY (rabbit LD₅₀): > 5,000 mg/kg

INHALATION TOXICITY (rat LC₅₀): > 4.24 mg/L (4-hour)

EYE IRRITATION: Rabbit – slightly irritating

SKIN IRRITATION: Rabbit – essentially non-irritating

SKIN SENSITIZATION: Guinea Pigs - Not a contact sensitizer

CARCINOGENICITY:

EPA: Not Listed

ACGIH: Not Listed

NTP: Not Listed

IARC: Group 2A¹

OSHA: Not Listed

MUTAGENIC TOXICITY: No evidence of mutagenic effects during *in vivo* and *in vitro* assays.

REPRODUCTIVE TOXICITY: Reproductive effects in rats only in the presence of significant maternal toxicity.

¹ Carcinogenicity: Not carcinogenic in rats or mice. Listed as Category 2A by the International Agency for Research on Cancer (IARC) but, in the expert opinion of the industry, this classification as a carcinogen is not warranted.

12. ECOLOGICAL INFORMATION

The following information is for the active ingredient, Glyphosate IPA salt:

AQUATIC TOXICITY:

Rainbow Trout and Bluegill (96-hr LC₅₀): > 1000 mg/L

Fathead minnow (96-hr LC₅₀): 97 mg/L

Daphnia (48-hr LC₅₀): 930 mg/L

Algae (72-hr EC₅₀): 72.9 mg/L

AVIAN TOXICITY:

Bobwhite quail: (oral LD₅₀) > 3,851 mg/kg-bw; (dietary LC₅₀) > 4,640 mg/kg-bw

Mallard duck: (dietary LC₅₀) > 4,640 mg/kg-bw

OTHER TOXICITY:

Honeybee: (contact LD₅₀) 100 µg/bee; (oral LD₅₀) > 100 µg/bee

13. DISPOSAL CONSIDERATIONS

PESTICIDE DISPOSAL: Wastes resulting from the use of this product must be disposed of on site or at an approved waste disposal facility.

CONTAINER DISPOSAL: Nonrefillable container. Do not reuse or refill this container. Refer to the product label for specific container disposal instructions.

14. TRANSPORT INFORMATION

US DOT:

Not regulated

IMDG:

Not regulated

IATA:

Not regulated

This information is not intended to convey all specific regulatory or operational requirements/information relating to this product. Transportation classifications may vary by container volume and may be influenced by regional or country variations in regulations. Additional transportation system information can be obtained through an authorized sales or customer service representative. It is the responsibility of the transporting organization to follow all applicable laws, regulations and rules relating to the transportation of the material.

15. REGULATORY INFORMATION

FIFRA –

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. The following is the hazard information as required on the pesticide label:

PRECAUTIONARY STATEMENTS

HAZARDS TO HUMANS AND DOMESTIC ANIMALS

Caution. Remove and wash contaminated clothing before reuse.

See inside label booklet for additional Precautionary Statements and Directions for Use including Storage and Disposal instructions.

ENVIRONMENTAL HAZARDS

Do not apply directly to water, to areas where surface water is present or to intertidal areas below the mean high water mark. Do not contaminate water when cleaning equipment or disposing of equipment washwaters and rinsate.

PHYSICAL OR CHEMICAL HAZARDS

Spray solutions of this product must be mixed, stored and applied using only stainless steel, aluminum, fiberglass, plastic or plastic-lined steel containers.

DO NOT MIX, STORE OR APPLY THIS PRODUCT OR SPRAY SOLUTIONS OF THIS PRODUCT IN GALVANIZED STEEL OR UNLINED STEEL (EXCEPT STAINLESS STEEL) CONTAINERS OR SPRAY TANKS. This product or spray solutions of this product react with such containers and tanks to produce hydrogen gas which may form a highly combustible gas mixture. This gas mixture could flash or explode, causing serious personal injury, if ignited by open flame, spark, welder's torch, lighted cigarette or other ignition source.

All pesticides are governed under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA). The regulatory information presented below is pertinent only when this product is handled outside of the normal use and application as a pesticide. This product is excluded from listing requirements under EPA/TSCA.

SARA Title III – Section 302 Extremely Hazardous Substances

Not listed

SARA Title III – Section 311/312 Hazard Categories

Immediate (Acute)

SARA Title III – Section 312 Threshold Planning Quantity

N/A

SARA Title III – Section 313 Reportable Ingredients

None

CERCLA Reportable Quantity (RQ) –

None

CALIFORNIA PROP 65 STATUS –

 **WARNING:** This product can expose you to chemicals including glyphosate, which is known to the State of California to cause cancer. For more information go to www.P65Warnings.ca.gov.

CANADA –

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR.

16. OTHER INFORMATION

THIS INFORMATION IN THIS SDS IS BASED ON DATA AVAILABLE AS OF THE REVISION DATE GIVEN HEREIN, AND BELIEVED TO BE CORRECT.

To the extent consistent with applicable law, Rainbow Agrosiences, LLC warrants that this product conforms to the chemical description on the label and is reasonably fit for the purposes set accordance with those Directions under the conditions described therein. NO OTHER EXPRESSED WARRANTY OR IMPLIED WARRANTY OF FITNESS FOR PARTICULAR PURPOSE OR MERCHANTABILITY IS MADE. This warranty is also subject to the conditions and limitations stated herein.

Buyer and all users shall promptly notify this Company of any claims whether based in contract, negligence, strict liability, tort or otherwise.

To the extent consistent with applicable law, buyer and all users are responsible for all loss or damage from use or handling which results from conditions beyond the control of the Company, including, but not limited to, incompatibility with products other than those set forth in the Directions, applications to or contact with desirable vegetation, unusual weather, weather conditions which are outside the range considered normal at the application site and for the time period when the product is applied, as well as weather conditions which are outside the application ranges set forth in the Directions, application in any manner not explicitly set forth in the Directions, moisture conditions outside the moisture range specified in the Directions, or the presence of products other than those set forth in the Directions in or on the soil, crop or treated vegetation.

To the extent consistent with applicable law, this Company does not warrant any product reformulated or repackaged from this product except in accordance with this Company's stewardship requirements and with express written permission from this Company.

TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, THE EXCLUSIVE REMEDY OF THE USER OR BUYER, AND THE LIMIT OF THE LIABILITY OF THIS COMPANY OR ANY OTHER SELLER FOR ANY AND ALL LOSSES, INJURIES OR DAMAGES RESULTING FROM THE USE OR HANDLING OF THIS PRODUCT (INCLUDING CLAIMS BASED IN CONTRACT, NEGLIGENCE, STRICT LIABILITY, OTHER TORT OR OTHERWISE) SHALL BE THE PURCHASE PRICE PAID BY THE USE OR BUYER FOR THE QUANTITY OF THIS PRODUCT INVOLVED, OR, AT THE ELECTION OF THIS COMPANY OR ANY OTHER SELLER, THE REPLACEMENT OF SUCH QUANTITY, OR, IF NOT ACQUIRED BY PURCHASE, REPLACEMENT OF SUCH QUANTITY IN NO EVENT SHALL THIS COMPANY OR ANY OTHER SELLER BE LIABLE FOR ANY INCIDENTAL, CONSEQUENTIAL OR SPECIAL DAMAGES.

To the extent consistent with applicable law, upon opening and using this product, buyer and all users are deemed to have accepted the terms of this LIMIT OF WARRANTY AND LIABILITY which may not be varied by any verbal or written agreement. If terms are not acceptable, return at once unopened.

SDS Version: 1.0

Effective Date: 09/23/2025